Analysis Report of Evaluation of Wisdomnet for Error-free ML

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1. Executive Summary

Today's modern machine learning classification systems frequently introduce prediction errors that are unacceptable in different types of mission critical applications. In order to solve the following, this report/analysis will provide "trustable learning" basically a paradigm where an artificial intelligence model avoids incorrect guessing rather than identifying uncertain data points and outputting a "reject" classification in order to ultimately leave the decision to human experts, WisdomNet is a novel neural network framework that attaches a zero initialized "conjugate neuron" to a pre-trained base model and helps fine-train it exclusively on the base model's past mistakes. Different results show and prove that WisdomNet consistently drops classification error rates to 0% across different architectures while successfully bounding human-intervention.

2. Introduction & Problem Statement

Introduction: We know that the standard supervised machine learning models helps optimize the overall classification accuracy, However in life-or-death work, we need absolute reliability before anyone can trust them

Problem: even the top AI models are not perfect, in life-or-death situations such as healthcare, medicine and driving even a small mistake can lead to disastrous outcomes, Human reviews are slow, and adding an "unkown" label creates more problem due to unbalanced data

3. Core Methodology: Trustable Learning & The WisdomNet Framework

Practicing trustable learning can help solve the problem by having AI flag uncertain cases and send them to a human expert. This helps ensure that zero errors occur in the real-world use

How does it work? (WisdomNet 4 Steps)

1. Train an AI model on regular data
2. Find mistakes: the AI will pick out its wrong predictions and group them under a

new “reject” label.

3. Add a new neuron: a special unit is attached to the AI’s output, starting from zero so it doesn’t change anything yet.
4. In case of mistakes, retrain the model: the new neuron is fine-tuned only on the past errors so it can catch the unstable or “lucky” guesses.

4. Testing and Real World Results

What we found:

1. No more errors: WisdomNet brought error rates down by 0% by sending uncertain cases to humans.
2. Work with any AI: It was tested on simple and the complex neural networks(MLP,ConvNet,and ResNet50) and worked on all of them.
3. Increased human effort: Only about 1 out of 10 cases needed a human check, so the system stays efficient.

5. Key Operational Insights

WisdomNet’s success depends upon a strong solid base model - below 90% accuracy, the contribution of human review begins at 30% - 50%. The upside to noisy data is that it makes the system more conservative, but it doesn’t cause it to make mistakes, And crucially, training is not all about learning from errors. Its about changing the way the model thinks, so that it detects unreliable predictions at source

6. The Hard Parts and Future Decision

While WisdomNet works really well in its current form, scaling up brings challenges. Its complexity is increased and the multi-category tasks overwhelm a single neuron, and even multi-neuron setups struggle with massive datasets. Also the training process needs a smarter and proper stopping mechanism in order to avoid over-rejection. These are the next big problems to solve.

7. Executive Summary

It's important to understand that we should not fully trust AI, as even the big stage LLM’s can still make mistakes using them in critical fields is dangerous. WisdomNet fixes this by flagging uncertain cases for humans- dropping errors to 0% while keeping human review under 10%. It works across all major AI types, but it still requires a strong base model and still struggles with complex AI problems.Overall, it's a promising step towards AI revolution